

$$A_1(n) = 2n - 1 = 2n^1 - 1$$

$$A_2(n) = 4n^2 - 4n - 5 = 4n^2 - 5$$

$$A_3(n) = 8n^3 - 12n^2 - 30n + 1 = 8n^3 + 2n^2 - 34n^1 + 1$$

$$A_4(n) = 16n^4 - 32n^3 - 120n^2 + 8n + 41 = 16n^4 + 64n^3 - 104n^2 - 128n^1 + 41$$

$A_t(n)$ is an Appel Sequence

$$\hookrightarrow \frac{d}{dx} p_n(x) = n p_{n-1}(x)$$

maybe a different constant

PROVE

$$g_t(n) = \frac{A_t(n)}{2^t t!} \Rightarrow g'_t = g_{t-1}$$

Conjectures

$$A_t(n) = \sum_{r=0}^t 2^{t-r} \binom{t}{r} \underbrace{A_r(0)}_{\substack{\text{constant} \\ \text{"unknown"}}} n^{t-r}$$

$$f_t(n) = \frac{A_t(n)}{2^t} \Rightarrow f'_t(n) = t f_{t-1}(n)$$

This is the constant term also referenced as $c_t(n)$.

Coming up:

Prove $A_t(n) = \text{empirical fits}$

Prove normalized Appelle series

Wrap up loose ends and investigate $A_t(0)$ terms